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S. Korea Technology | Asia Pacific

Asia Summer School: AI, Memory and MLCC

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S. KOREA TECHNOLOGY

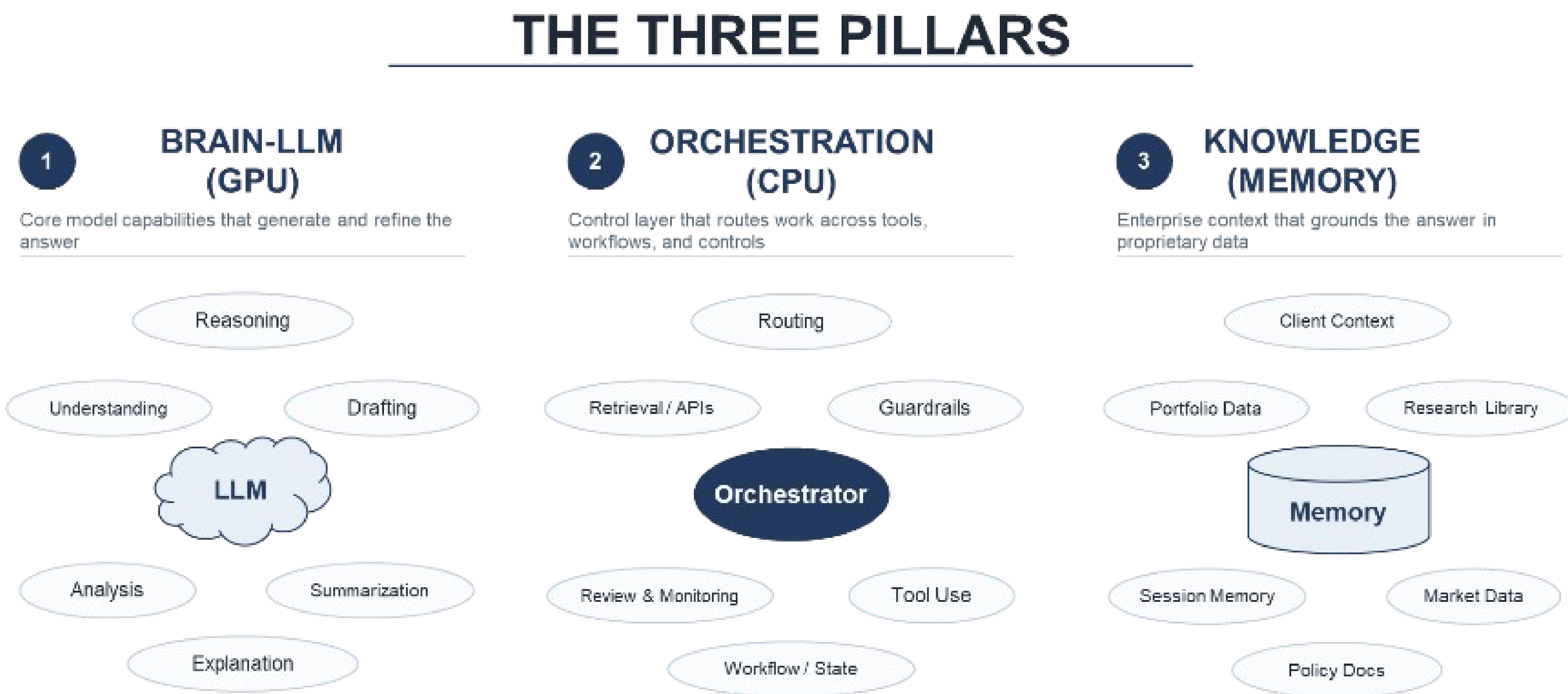
Asia Pacific

Industry View **Attractive**

Asia Summer School 2026

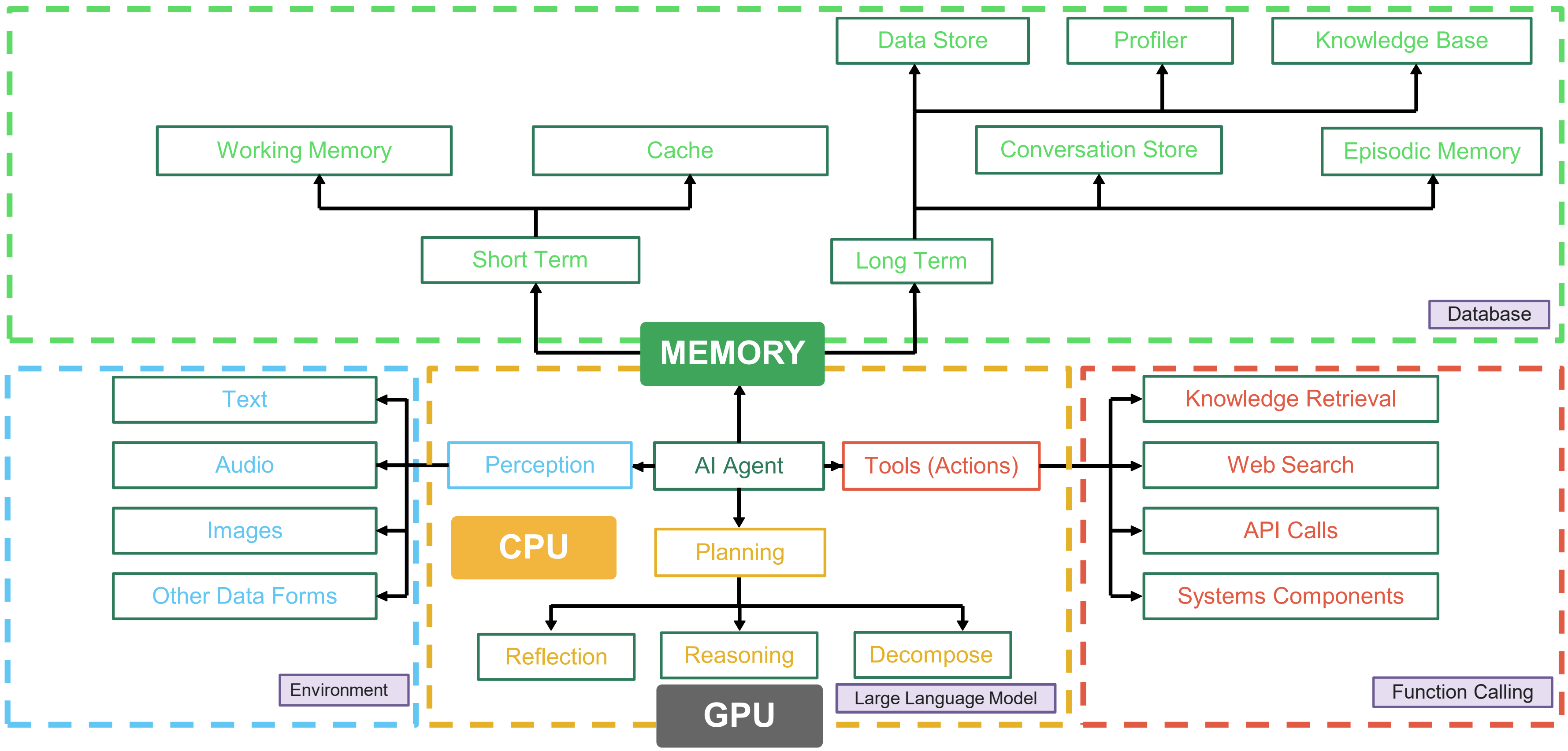
Rise of the AI Agent

AI Transitions from "Generation" to "Autonomous" Action in Agentic AI



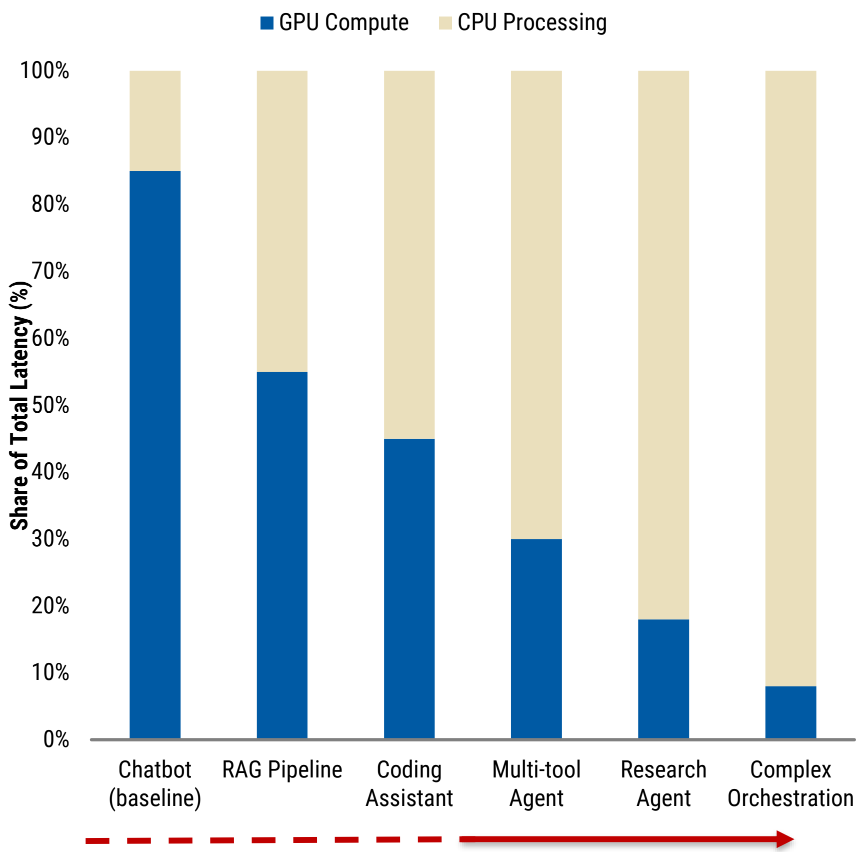
Source: Morgan Stanley Research.

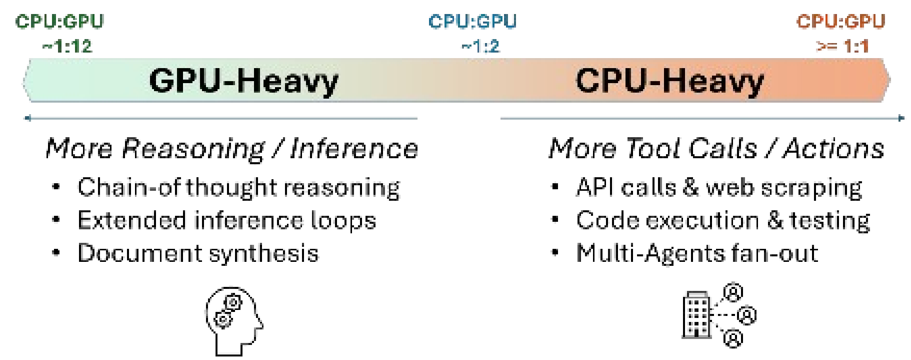
AI Agent Architecture – CPU Orchestration, Memory and GPU Execution



Source: Mongo DB.

CPU Is the New Bottleneck in Agentic AI

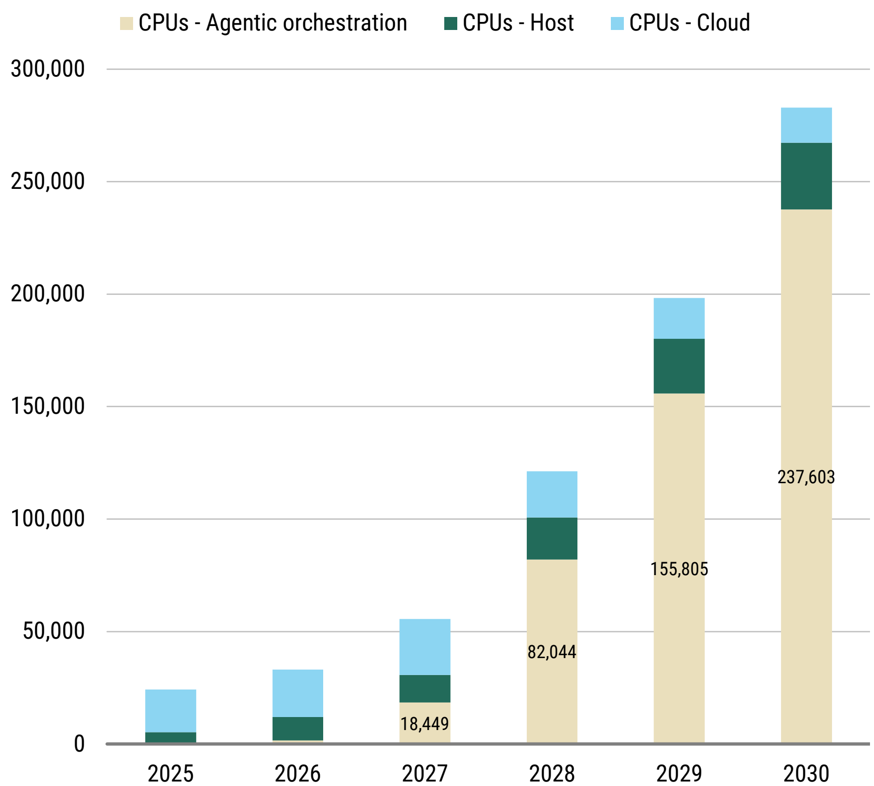


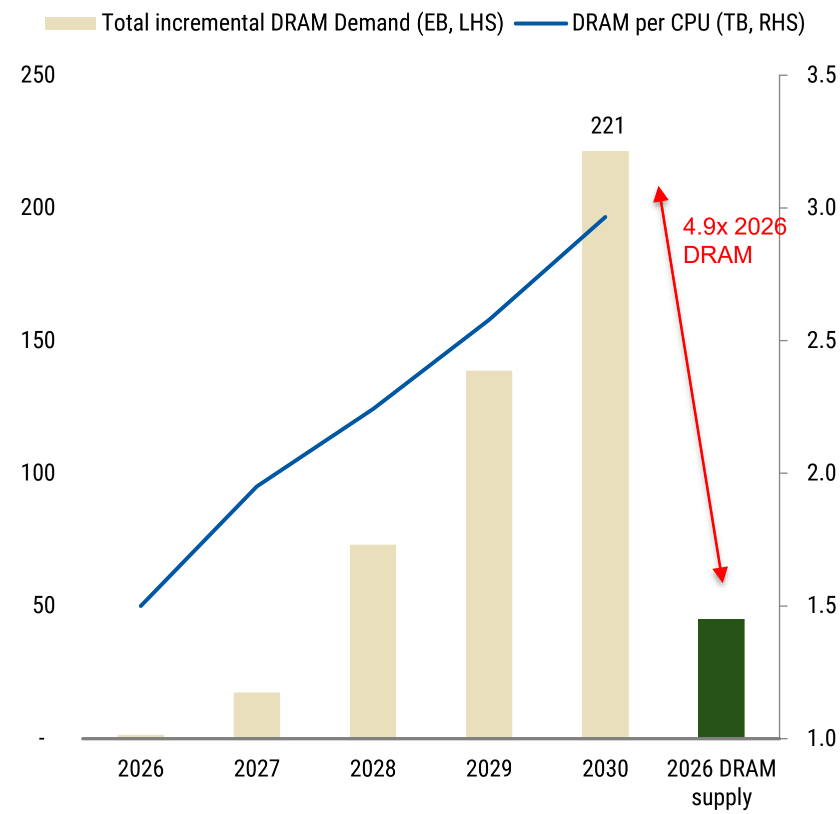


Source: Morgan Stanley Research, Georgia Tech and Intel paper. Note: Per-workload splits are our own directional estimates, not measured benchmarks.

Agentic AI Opportunity – Sizing the CPU and Memory TAM

- Agentic AI could add up to US$238bn CPU opportunity and 221EB of DRAM demand by 2030 under our bull-case scenario





Source: Morgan Stanley Research estimates.

Exposure to the AI Agent Theme

Global Exposure Across the Stack ? Names by exposure

CPU	DRAM	NAND	HDD	FOUNDRY	IC-design
• NVDA • Intel • AMD • Arm	• Samsung • Hynix • Micron	• Kioxia • SanDisk	• Seagate • WDC • TDK	• TSMC	• GUC • Egis

PCB/Substrate/CCL & Materials

- SEMCO • Unimicron • NYPCB
- Ibiden • Nittobo • MEC

BMC, CPU & Memory interface

- Aspeed • Renesas • Montage
- WPG • AP Memory

MLCC & CPU socket

- Murata • TDK • Yageo
- FIT Hon Teng • Lotes

ODM

- Wiwynn
- Hon Hai

SPE

- ASML • ASMi • AMAT • Besi • KLAC
- Tokyo Electron • Ulvac • Wonik

Source: Morgan Stanley Research.

AI Investment Stack – Value Moving From Hardware to Applications

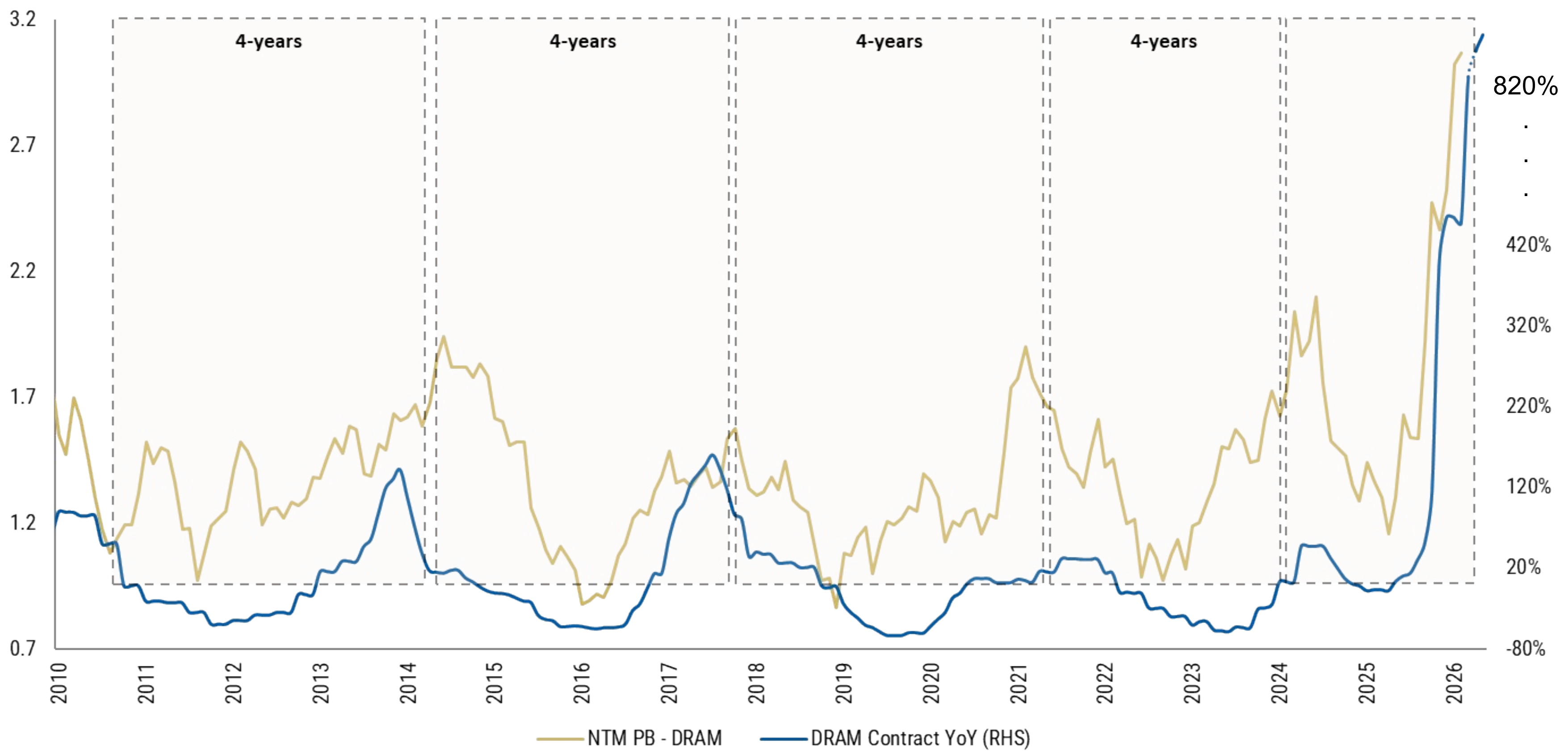
Source: Morgan Stanley Research.

AI chip value chain: China vs. the US – A Decoupling of AI Compute

Source: Gartner, Morgan Stanley Research (e) estimates.

Memory Update

Memory Cycles Are Supply-driven "Rinse & Repeat"



Source: FactSet, DRAMxChange, Morgan Stanley Research.

Memory: Changing Tides – Three Debates Framing the Cycle

How we frame the memory debate for investors – where the money is being spent, how the market prices LTAs, and where the cycle now sits.

1 AI Spend

+

CONSTRUCTIVE

Hyperscale capex spend

- Intensity of top LLMs
- Funding
- ARR (Annual Revenue Run Rate)

Monetizing infrastructure ≠ excess compute; 2Q26 capex is the tell

2 LTA Re-rating

=

NEUTRAL

No re-rating on LTAs

- Analog case study during COVID
- LTAs renegotiated / forced inventory
- Structural vs. cyclical debate

Market is pricing memory earnings rationally, not euphorically

3 Memory Cycle

=

NEUTRAL

Inflection

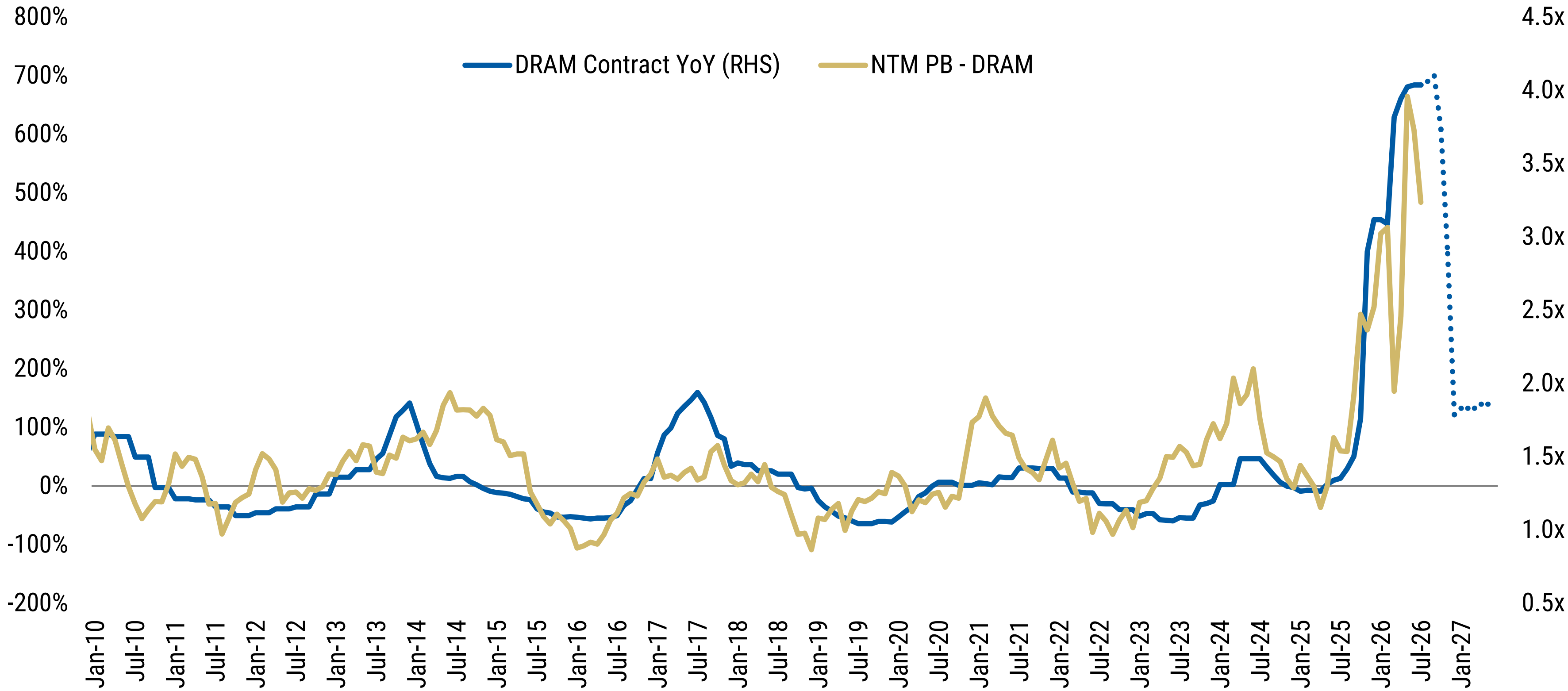
- Δ YoY rate of change
- Inventory
- “Good news = bad news”

Peaking rate of change, but the cycle elongates rather than collapses

Source: Morgan Stanley Research

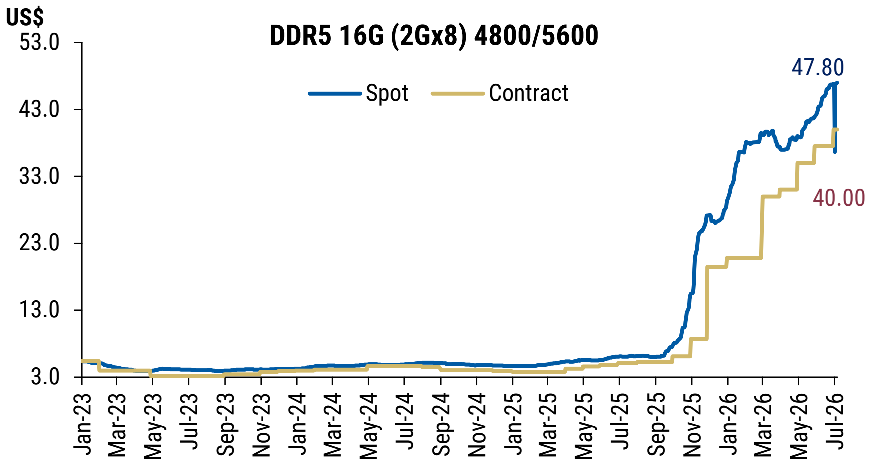
Memory Pricing Likely To Peak in 4Q26

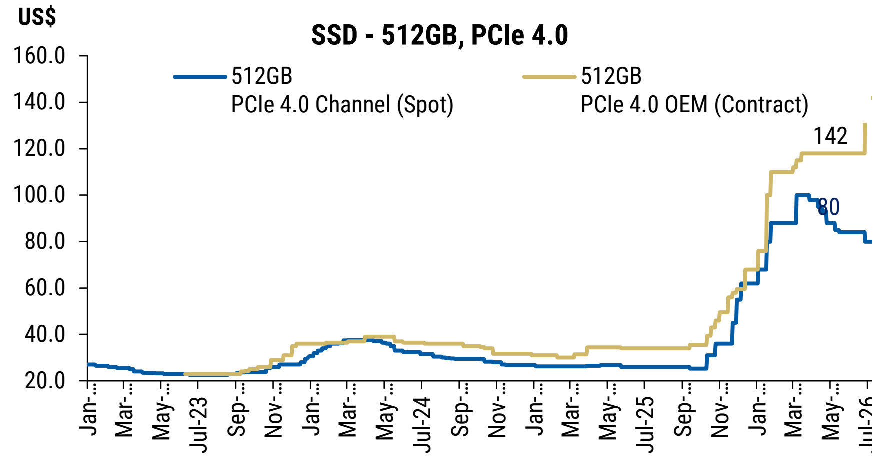
DRAM contract price YoY is rolling over from cyclical peaks, while valuation (NTM P/B) has yet to re-rate – consistent with pricing peaking around 4Q26.

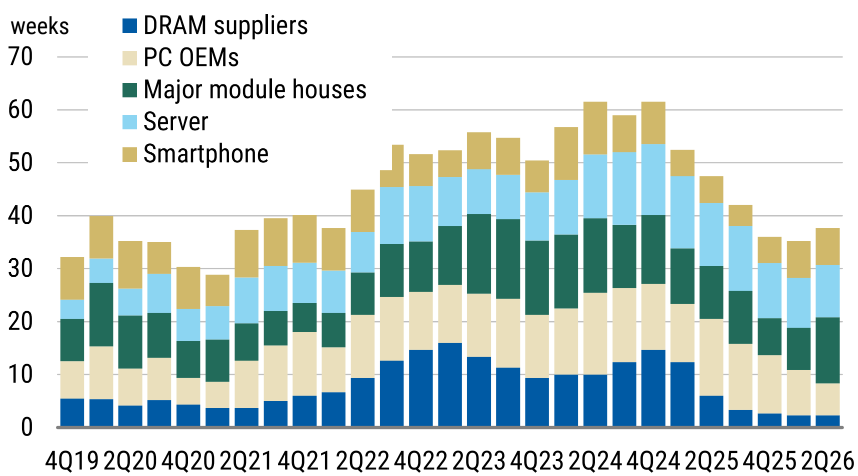


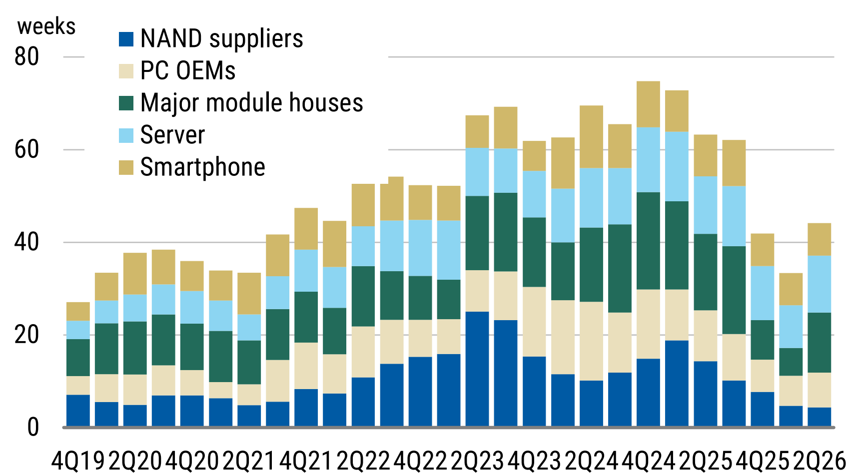
Source: TrendForce, FactSet, Morgan Stanley Research. Note: dot line is MS forecast.

Contract and Spot Pricing – Unprecedented Mismatch







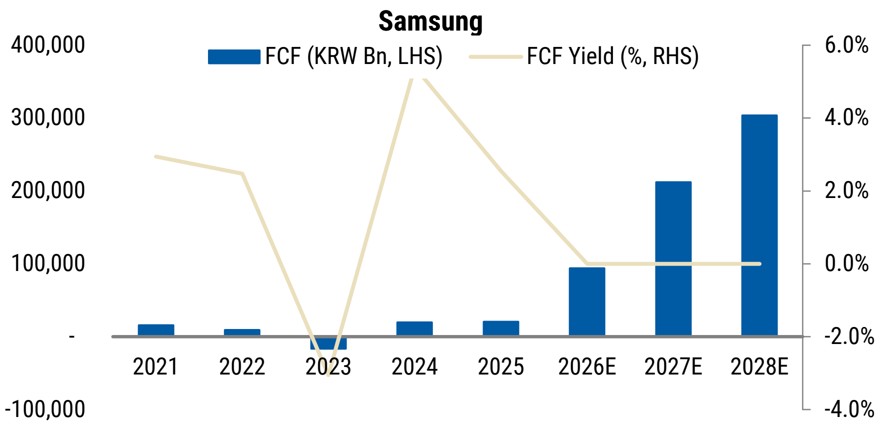


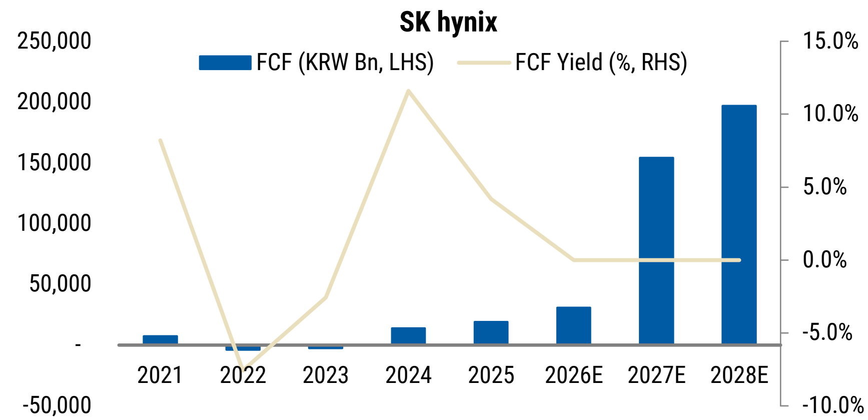
LTAs Framework Estimates – Old vs. New

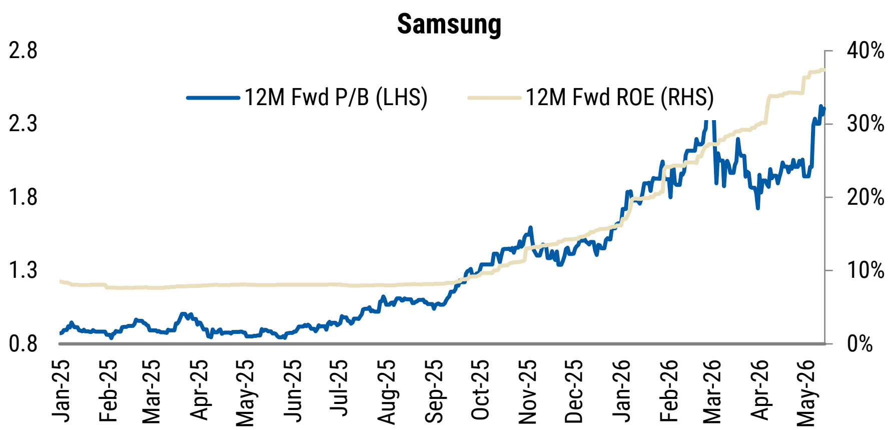
Feature	Traditional LTAs	Current LTA Frameworks
Duration	Quarterly, annual, or soft multi-year	Typically 3-5 years
Volume commitment	Often flexible / best-effort	Pre-agreed supply allocation or volume commitment
Pricing	Frequently renegotiated with the cycle	Pricing formula, range, floor, collar, or fixed mechanism
Customer commitment	Limited penalty / weak binding power	Prepayment, collateral, financial guarantee, or stronger binding terms in some agreements
Supplier benefit	Some demand visibility	Revenue visibility, margin protection, upfront capital, improved capacity planning
Customer benefit	Allocation during shortage	Strategic supply security for AI infrastructure buildout
Investor relevance	Limited valuation impact	Potential P/E re-rating if earnings become more contracted and less cyclical

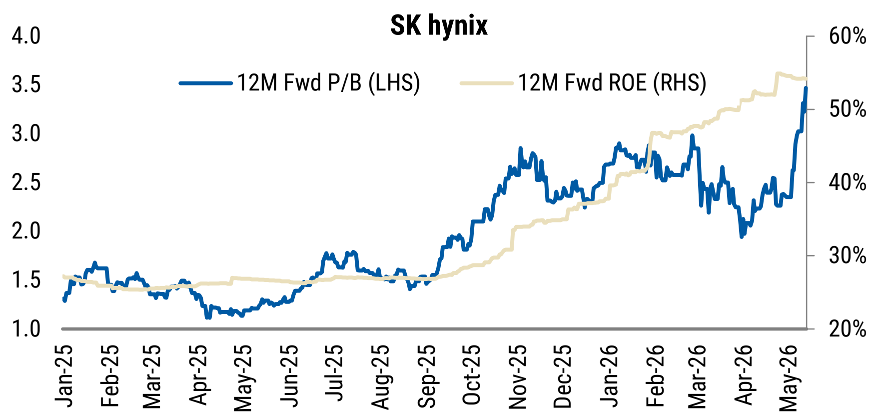
Source: Morgan Stanley Research estimates.

LTAs Drive Sustained FCF … And Hence Shareholder Returns and Re-rating





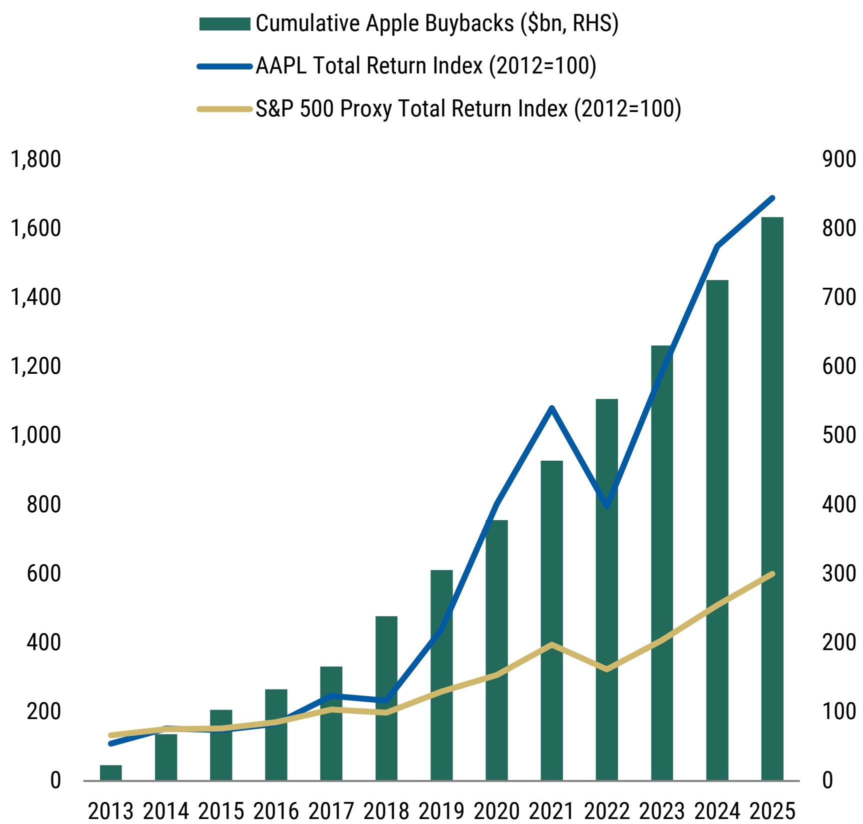


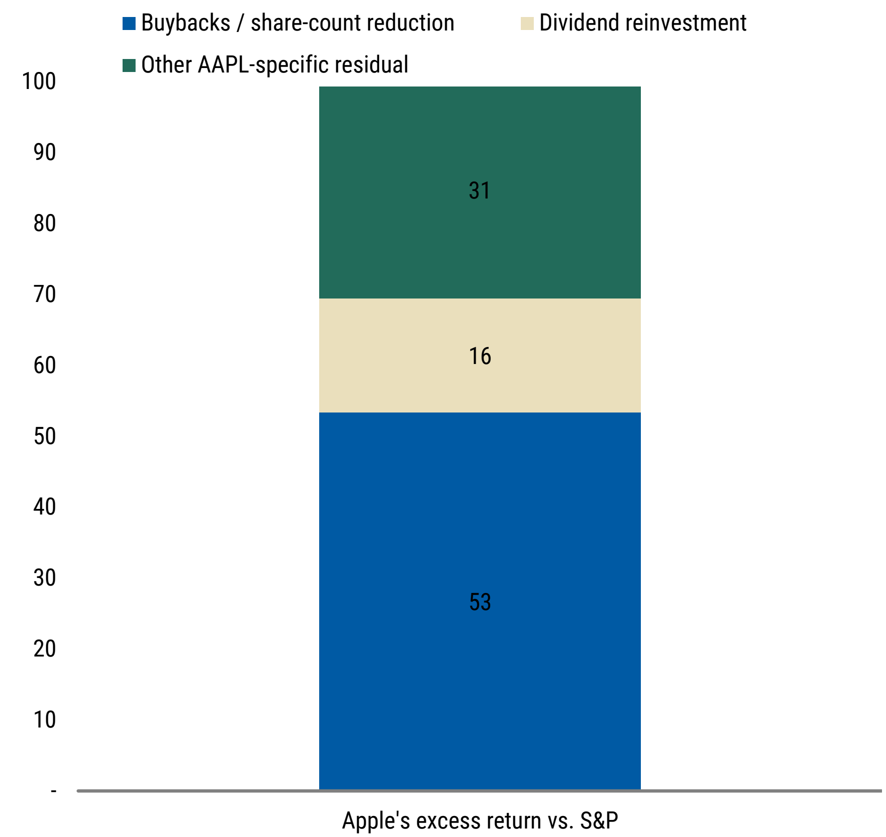


Source: Morgan Stanley Research estimates.

Apple Stock Provides a Useful Playbook of What's To Come

- Apple’s buybacks and dividends drove ~70% of excess compounded return vs. S&P proxy during 2013-25

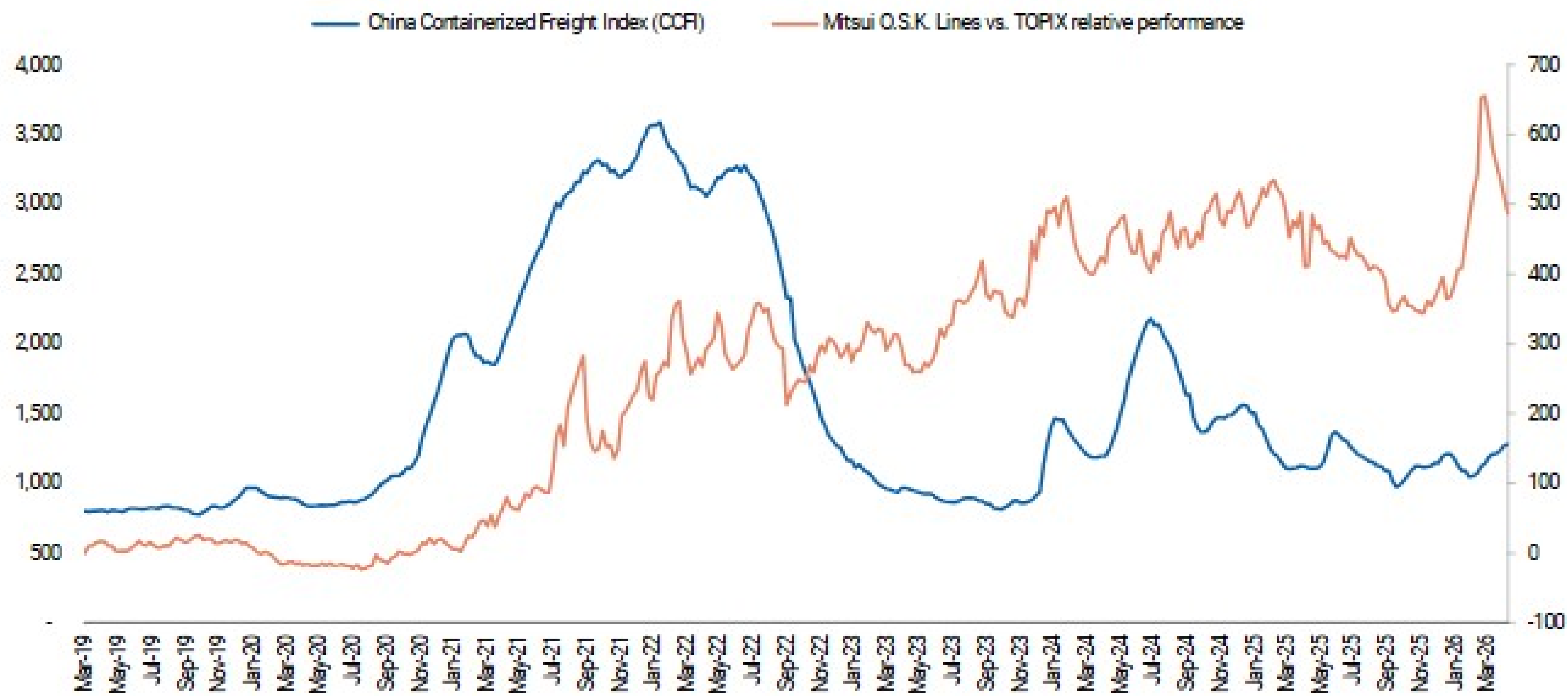




Source: Morgan Stanley Research estimates.

Japanese Shipping Companies Are Another Example of How Memory Valuation Can Re-rate

- Continued outperformance from enhanced dividend payout ratio after Covid



Source: Morgan Stanley Research estimates.

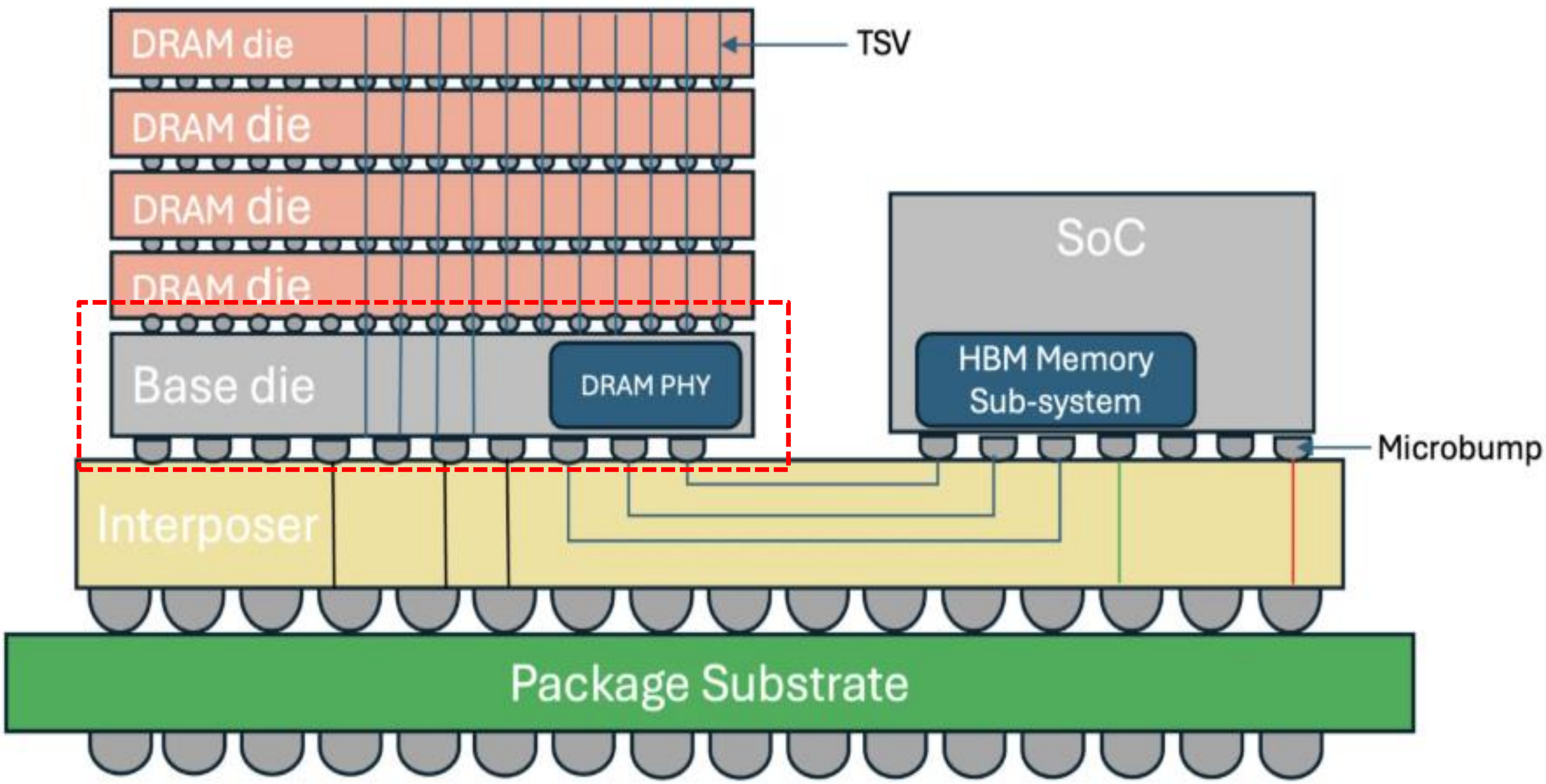
Quantifying LTAs

Samsung – implied group P/E sensitivity							
Commodity LTA % down; LTA P/E across							
	6.0x	**7.0x**	**8.0x**	**9.0x**	**10.0x**	**11.0x**	**12.0x**
50%	5.5x	6.0x	6.5x	7.0x	7.6x	8.1x	8.6x
60%	5.6x	6.2x	6.8x	7.4x	8.0x	8.6x	9.2x
70%	5.7x	6.4x	7.1x	7.8x	8.5x	9.2x	9.9x
80%	5.8x	6.6x	7.4x	8.1x	8.9x	9.7x	10.5x
Samsung current FY27 P/E	**3.8x**						

SK hynix – implied group P/E sensitivity							
Commodity LTA % down; LTA P/E across							
	6.0x	**7.0x**	**8.0x**	**9.0x**	**10.0x**	**11.0x**	**12.0x**
50%	5.5x	6.1x	6.6x	7.2x	7.7x	8.3x	8.8x
60%	5.6x	6.3x	6.9x	7.5x	8.2x	8.8x	9.5x
70%	5.7x	6.5x	7.2x	7.9x	8.6x	9.4x	10.1x
80%	5.8x	6.6x	7.5x	8.3x	9.1x	9.9x	10.7x
SK hynix current FY27 P/E	**3.9x**						

Source: Morgan Stanley Research estimates.

HBM Logic Base Die Diagram



Source: SemiWiki.

HBM TAM

Base case	2023	2024	2025	2026e	2027e	23-27e CAGR
NVIDIA calculation						
GP GPU volume (k units)	1,474	3,609	6,359	9,565	11,478	*67%*
NVIDIA market share (%)	*90%*	*90%*	*88%*	*92%*	*90%*	
Total GPU volume (k units)	1,638	4,010	7,226	10,397	12,753	*67%*
ASIC calculation						
Total ASIC volume (k units)	2,556	5,298	5,053	5,951	9,819	*40%*
Y/Y %		107%	-5%	18%	*65%*	
HBM calculation						
HBM usage per GPGPU (GB)	80	143	208	288	317	*41%*
HBM usage per ASIC (GB)	40	70	141	150	238	*56%*
HBM ASP per GB (US$)	15	12	13	13	13	*-3%*
Inventory Buffer		20%	10%	10%	10%	
Total HBM Usage (mn GB)	233	1,132	2,440	4,276	7,011	*134%*
Y/Y %		*385%*	*115%*	*75%*	*64%*	
Total HBM market (US$bn)	**3**	**13**	**31**	**57**	**94**	*128%*
Total HBM market (mn Gb)	**1,866**	**9,059**	**19,516**	**34,205**	**56,085**	

Source: Company data, Morgan Stanley Research (e) estimates. Note: Estimates are compiled using our Asia supply-chain checks.

HBM Sufficiency Estimates

Sufficiency estimates	2023	2024	2025	2026e	2027e
HBM TSV Capacity (K wpm)					
Samsung	45	130	150	180	250
SK Hynix	45	120	150	200	250
Micron	3	20	60	100	110
Total	**93**	**270**	**360**	**480**	**610**
Yield rate assumptions					
Samsung		*50%*	*60%*	*55%*	*70%*
SK Hynix		*60%*	*75%*	*65%*	*70%*
Micron		*50%*	*70%*	*65%*	*70%*
UTR assumptions					
Samsung		*80%*	*60%*	*70%*	*100%*
SK Hynix		*100%*	*100%*	*100%*	*100%*
Micron		*100%*	*100%*	*100%*	*100%*
Implied HBM production (mn Gb)					
Samsung	1,500	4,435	6,387	9,696	21,239
SK Hynix	1,500	6,273	12,830	17,363	22,226
Micron	150	729	3,548	7,937	10,372
Total	**3,150**	**11,436**	**22,765**	**34,997**	**53,837**
Sufficiency Ratio					
HBM Demand (mn Gb)	1,866	9,059	19,516	34,205	56,085
Commodity DRAM market demand (mn Gb)	209,527	243,423	316,640	469,675	619,970
Total HBM+DRAM demand (mn Gb)	211,393	252,481	336,156	503,879	676,055
Commodity DRAM market supply (mn Gb)	200,214	228,086	301,257	382,107	519,666
Total HBM+DRAM supply (mn Gb)	203,364	239,523	324,022	417,104	573,503
Total DRAM sufficiency	***-4%***	***-5%***	***-4%***	***-17%***	***-15%***

Source: Company data, Morgan Stanley Research (e) estimates. Note: Estimates are compiled using our Asia supply-chain checks.

AI NAND TAM

	2025	2026e	2027e
ASIC			
AWS Trainium (k units)	1,285	1,700	2,380
ASIC per compute tray	4	4	4
AWS Tray # (k units)	321	425	595
eSSD per tray (TB)	32	32	32
Total eSSD (EB)	***10***	***14***	***19***
Meta MTIA (k units)	50	150	550
ASIC per compute tray	4	4	4
MTIA Tray # (k units)	13	38	138
eSSD per tray (TB)	64	64	64
Total eSSD (EB)	***1***	***2***	***9***
Google TPU (k units)		4,080	8,168
TPU per compute tray		4	4
Google Tray # (k units)		1,020	2,042
eSSD per tray (TB)		16	16
Total eSSD (EB)		***16***	***33***
Total ASIC eSSD usage (EB)	***11***	***32***	***61***

	2025	2026e	2027e
GPGPU			
Nvidia (k units)			
B200/B300	5,819	5,460	560
R200		2,080	5,920
R300			1,040
In-rack eSSD			
GPU per compute tray	4	4	4
Total # of compute tray	1,455	1,885	1,880
eSSD per tray (TB)	18	18	18
Total eSSD (EB)	**26**	**34**	**34**
Blackwell extra (CSP deployment) per tray	46	46	46
Total eSSD (EB)	**67**	**63**	**6**
STX CMS Context Memory Platform			
# of NVL72 Rack		28,889	96,667
# of CMS vs NVL72 Rack		0.125	0.125
# of storage tray (unit)		16	16
eSSD per tray (TB)		32	32
Total eSSD (EB)		**2**	**6**
AMD (k units)			
MI300/MI308	180	36	-
MI325/350/375	540	84	288
MI400		650	1,920
MI500			96
GPU per compute tray	4	4	4
Total # of compute tray	180	193	576
eSSD per tray (TB)	64	64	64
Total eSSD (EB)	**12**	**12**	**37**
Total GPGPU eSSD usage (EB)	**105**	**111**	**83**
CPU Rack			
Vera CPU		1,030	2,270
eSSD per CPU (TB)		8	8
Total eSSD (EB)		**8**	**18**
Others			
China CSP usage (EB)	80	120	180
QLC usage (EB)	-	100	230
CSP Inventory buffer %	5%	10%	10%
Total AI NAND demand (EB)	**205**	**400**	**609**

Source: Company data, Morgan Stanley Research (e) estimates. Note: Estimates are compiled using our Asia supply-chain checks.

NAND Sufficiency Estimates

	2025	2026e	2027e
Total AI NAND demand (EB)	**205**	**400**	**609**
Other Applications (EB)			
non-AI SSD (general server, PC, enterprise etc)	396	387	401
PC SSD	209	181	175
Enterprise SSD (non-AI)	187	206	226
Smartphones	337	291	301
Tablets	34	34	34
Flash Cards	15	15	15
USB Flash Drives	10	10	10
Other	113	113	113
Total non-AI NAND demand (EB)	**905**	**850**	**874**
Total NAND demand (EB)	1,111	1,250	1,484
Total NAND supply (EB)	1,128	1,058	1,347
Sufficiency Ratio (%)	***2%***	***(15%)***	***(9%)***
AI as % of NAND demand (EB)	*18%*	*32%*	*41%*

Source: Company data, Morgan Stanley Research (e) estimates. Note: Estimates are compiled using our Asia supply-chain checks.

NAND: YMTC Supply vs. Demand Scenario Analysis

- **Capacity:** YMTC is building Fab4 and Fab5 (~100kwpm each); dedicating all five announced fabs to NAND could take it to ~24% of global NAND share.
- **Base case:** non-AI NAND demand +5% YoY and AI SSD demand +30–60% YoY in 2028, vs. YMTC capacity of 310kwpm (base) to 470kwpm (max).
- **Bottom line:** NAND stays tight into 2028 if AI capex keeps growing and YMTC stays disciplined; faster greenfield expansion is the key oversupply risk.

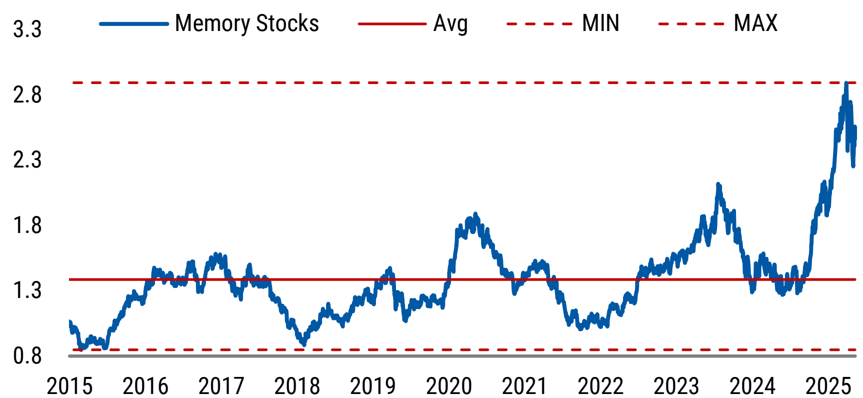
AI SSD demand growth % YoY in 2028

YMTC Capacity kwpm in 2028e	30%	40%	50%	60%
310	-1%	-2%	-4%	-6%
330	0%	-1%	-3%	-4%
350	2%	0%	-2%	-3%
370	3%	1%	-1%	-2%
390	4%	2%	1%	-1%
410	5%	4%	2%	0%
430	7%	5%	3%	1%
450	8%	6%	4%	3%
470	9%	7%	5%	4%

Source: FactSet, Morgan Stanley Research

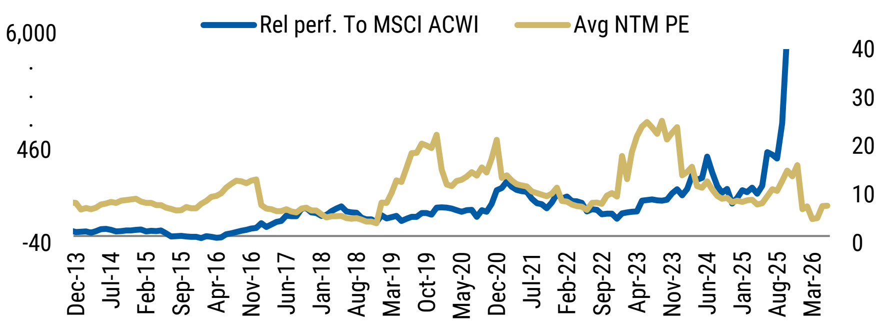
Trading in Memory Is Efficient

P/B Predict Inflections



P/E Less Relevant for Predicting Future Performance

Avg. DRAM Stock Relative perf. (left) vs. cons. forward P/E (right)



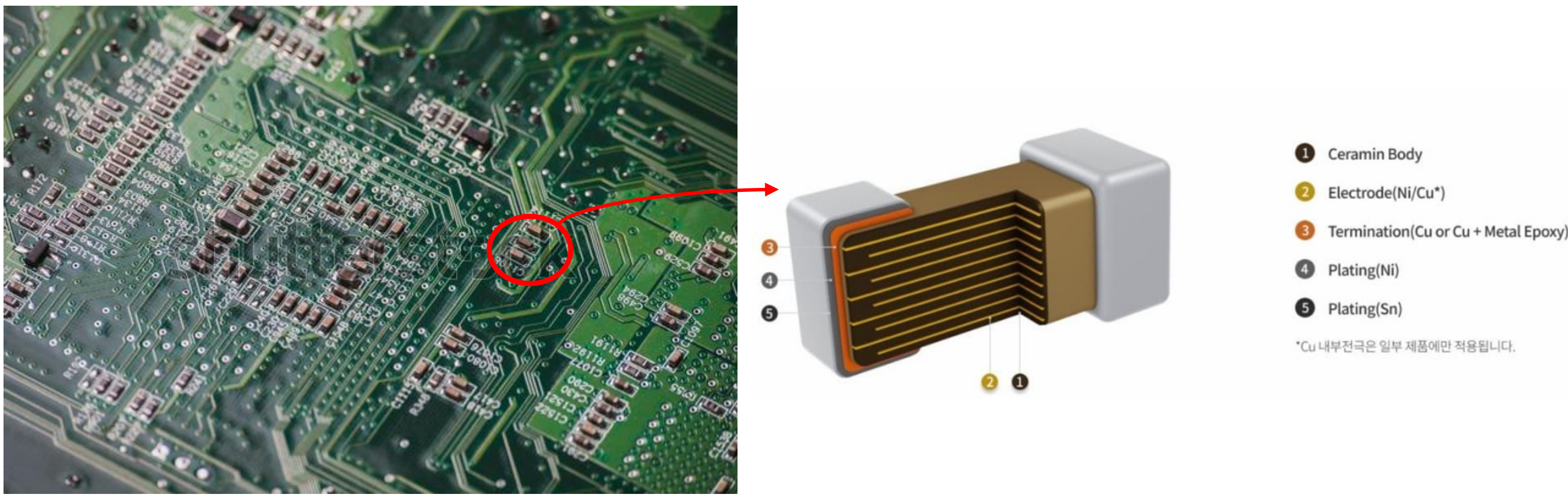
- **Buy when the group looks expensive; sell when everyone is bullish**.
- Avoid the group when stocks appear cheap – because estimates are coming down.
- Memory stocks tend to look cheap when the market is discounting a downturn well ahead of Street estimates.
- Street tends to miss chip inventory corrections at the peak or full-blown cyclical downturns.
- On the flip side, when memory revenues miss expectations, Street estimates get too pessimistic.
- Memory stocks, on average, currently trade on 2.5x P/B, which is above the historical average.
- P/B useful for trough or peak cycle multiples.

Source: MSCI, Morgan Stanley Research, Company data. Note: Avg performance of Samsung Electronics, Micron, and SK Hynix

MLCC Update

MLCCs: Small Components That Enable Modern Electronics

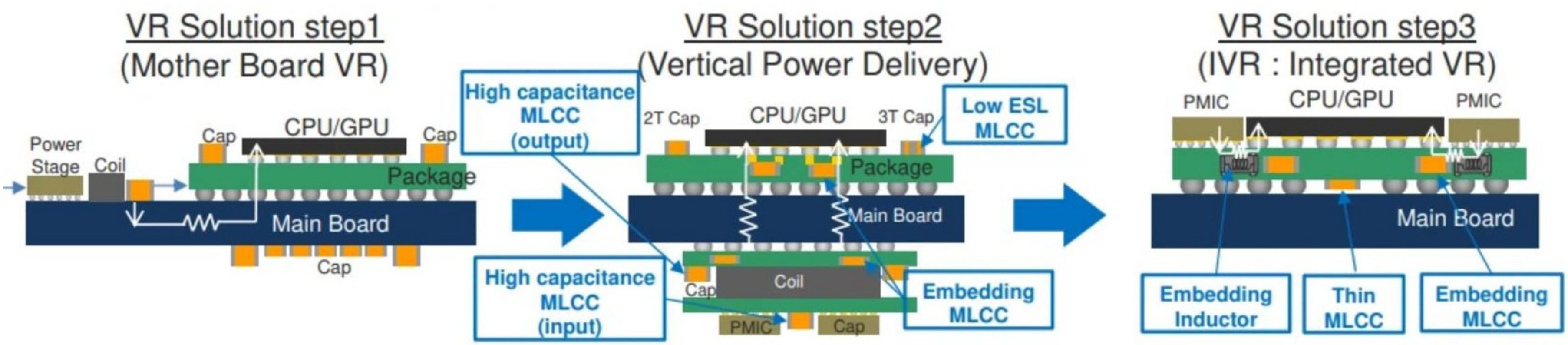
MLCCs (Multi-Layer Ceramic Capacitors) are essential passive components used for power stabilization, noise filtering, and signal integrity.



Source: Shutterstock, Samsung Electro-Mechanics

AI Servers Require More MLCCs as Power Delivery Architectures Evolve

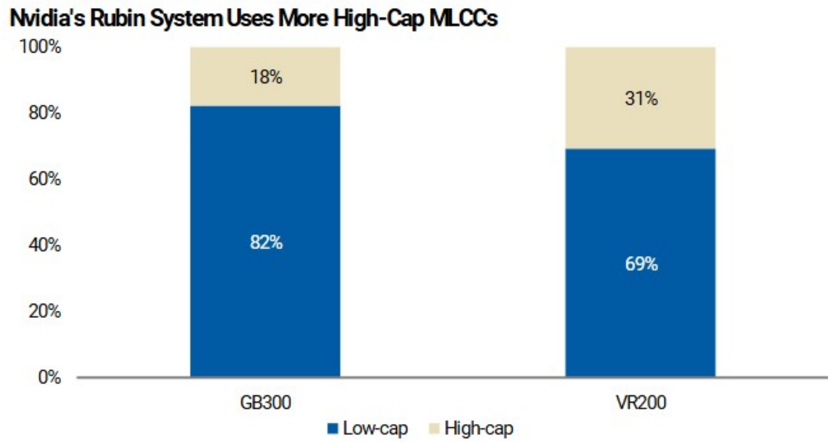
New server designs require higher-capacitance, lower-ESL, and embedded MLCC solutions positioned closer to CPUs and GPUs.



Source: Taiyo Yuden.

VR200 MLCC Content Is 180%+ Higher Compared to GB300

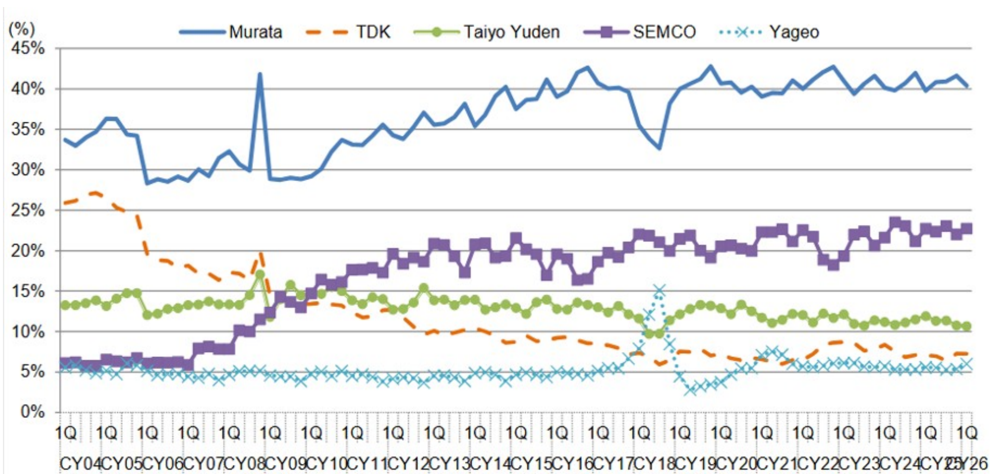
MLCC content per board (US$)	GB300	VR200
Compute PCB	$25	$90
Switch PCB	$20	$45
BlueField DPU Module	$5	$5
ConnectX Orchid Module	$5	$5
Other peripheral PCB	$5	$5
Units per rack	**GB300**	**VR200**
Compute PCB	36x	36x
Switch PCB	9x	9x
BlueField DPU Module	0x	18x
ConnectX Orchid Module	0x	72x
Other peripheral PCB	90x	45x
Total MLCC Content per rack	**GB300**	**VR200**
Compute PCB	$900	$3,240
Switch PCB	$180	$405
BlueField DPU Module	$0	$90
ConnectX Orchid Module	$0	$360
Other peripheral PCB	$450	$225
Total MLCC Content per rack	**$1,530**	**$4,320**

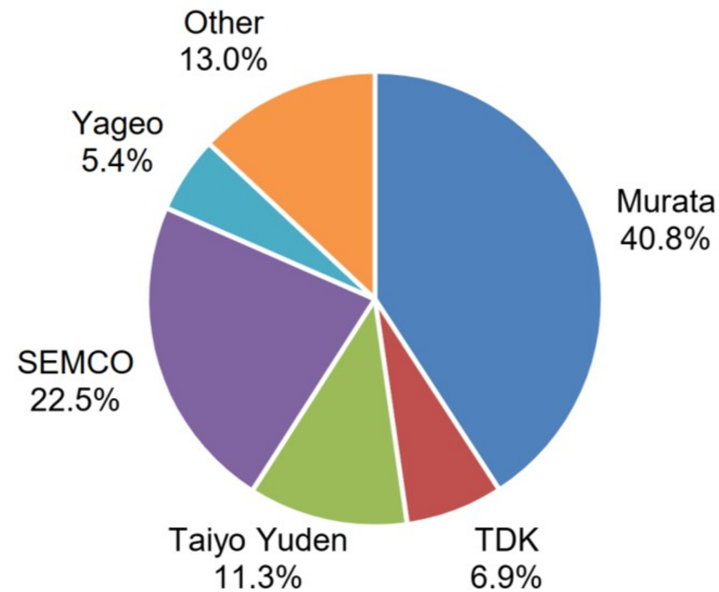


Source: Morgan Stanley Asia Pacific Research estimates based on supply chain checks. Note: High-cap refers to 47uF+ MLCCs.

The MLCC Industry Is Highly Consolidated

The top five suppliers account for ~87% of the global MLCC market (CY25), led by Murata and SEMCO.

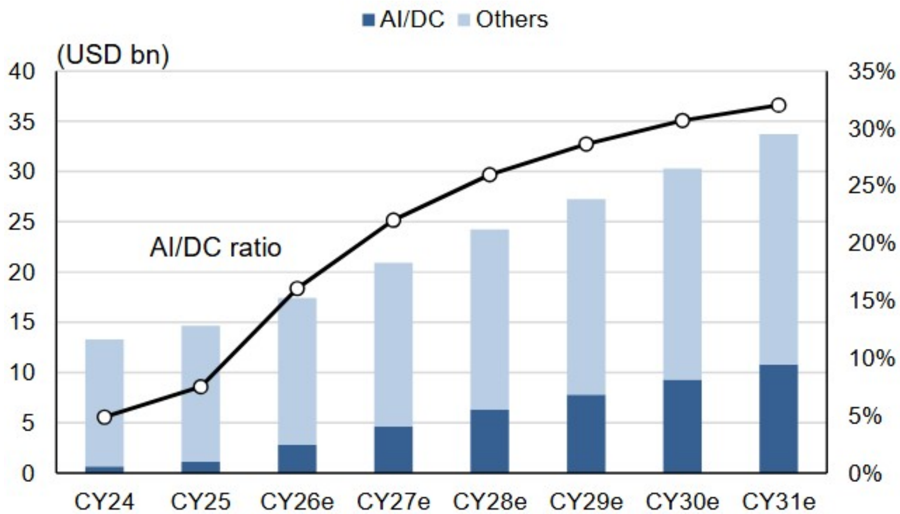


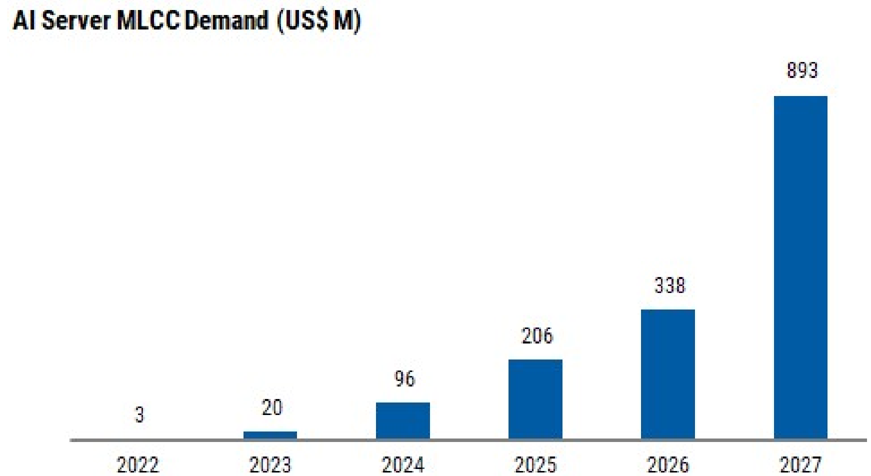


Source: Company data, Morgan Stanley Research.

AI Is Emerging as a Key Growth Driver for the MLCC Industry

We estimate AI server MLCC demand to approach US$1bn by 2027 driven by content growth and spec upgrade.

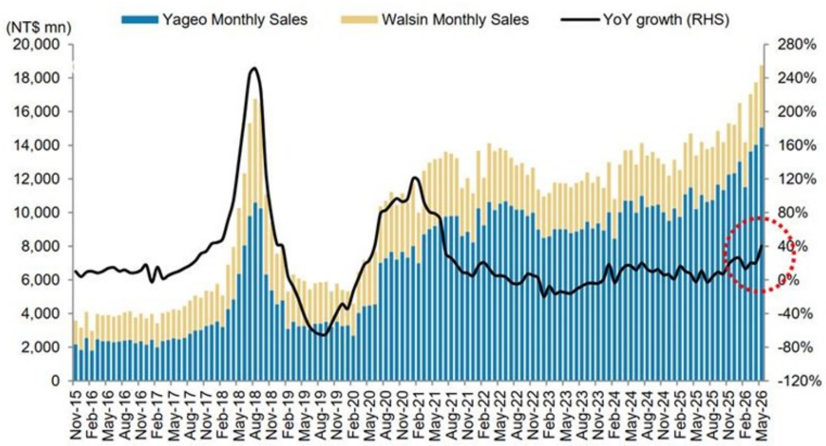


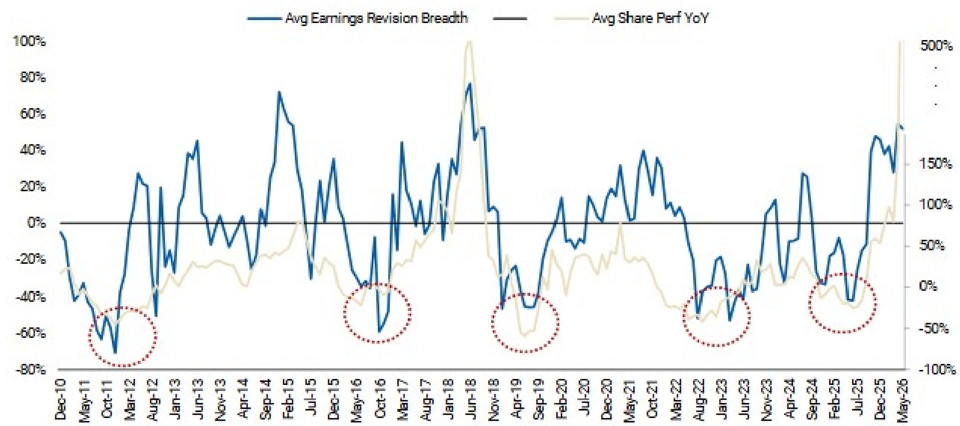


Source: Company data, Morgan Stanley Research estimates.

MLCC Fundamentals Are Inflecting

Monthly sales growth among key Taiwan MLCC suppliers has recently turned positive. Earnings revision breadth has improved meaningfully in recent months.





Source: FactSet, Morgan Stanley Research estimates.

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(as of June 30, 2026)

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	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
Stock Rating Category	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1544	42%	453	49%	29%	757	44%
Equal-weight/Hold	1577	43%	390	42%	25%	769	44%
Not-Rated/Hold	3	0%	1	0%	33%	1	0%
Underweight/Sell	544	15%	89	10%	16%	204	12%
Total	3,668		933			1731	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

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INDUSTRY COVERAGE: S. Korea Technology

COMPANY (TICKER)	RATING (AS OF)	PRICE* (07/14/2026)
Ryan Kim		
Ecopro BM (247540.KQ)	U (03/20/2023)	W112,600
Fadu Inc (440110.KQ)	O (09/21/2025)	W71,300
Hanmi Semiconductor Co. Ltd. (042700.KS)	O (08/16/2024)	W207,500
HD Hyundai Electric Co Ltd (267260.KS)	O (03/25/2025)	W797,000
Isu Petasys Co. Ltd. (007660.KS)	O (02/03/2025)	W97,900
L&F Co Ltd (066970.KS)	E (04/03/2025)	W90,800
Leeno Industrial Inc. (058470.KQ)	O (04/03/2025)	W73,100
Lotte Energy Materials Corp (020150.KS)	U (04/03/2025)	W31,450
LS Electric (010120.KS)	O (01/22/2026)	W187,700
POSCO FUTURE M (003670.KS)	U (04/03/2025)	W144,300
SK IE Technology (361610.KS)	U (04/03/2025)	W14,140
SK Square Co Ltd. (402340.KS)	O (05/06/2026)	W1,190,000
Wonik IPS Co Ltd (240810.KQ)	O (08/08/2025)	W127,100
Shawn Kim		
LG Display (034220.KS)	E (06/11/2025)	W10,350
LG Electronics (066570.KS)	E (04/07/2025)	W185,400
LG Innotek (011070.KS)	E (03/12/2025)	W641,000
Samsung Electro-Mechanics (009150.KS)	O (07/31/2025)	W1,260,000
Samsung Electronics (005935.KS)	O (11/18/2019)	W182,200
Samsung Electronics (005930.KS)	O (11/18/2019)	W263,000
SK hynix (000660.KS)	O (09/21/2025)	W1,913,000

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* Historical prices are not split adjusted.